Poles/Zeros & Stability

Control Engineering

Course Schedule

1. Systems
2. Continuous-time Controllers
3. Laplace Domain
4. Poles/Zeros & Stability
5. Controller Design
6. Discrete-time Controllers
7. State Space Representation
8. Advanced Control Methods

Poles/Zeros & Stability

- Effect on transient response
- Resonance
- Frequency-domain stability analysis

Poles and Zeros

$$G(s) = \frac{b_m}{a_n} \frac{\prod_{i=1}^{m} (s - s_{0i})}{\prod_{i=1}^{n} (s - s_{pi})}$$

Poles: roots of denominator s_p

Zeros: roots of numerator s_0

Determine system behavior

Interpretation of Poles

System's natural response:

sum of terms like $e^{s_p t}$ (one term for each pole $s_p = \sigma + j\omega$)

Real s_p: purely exponential, no oscillation

Complex s_p: (damped) oscillation

- Single real pole: PT1
- Two real poles: aperiodic PT2 (sum of two exponential terms)
- Conjugate pole pair: oscillating PT2

Pole Location

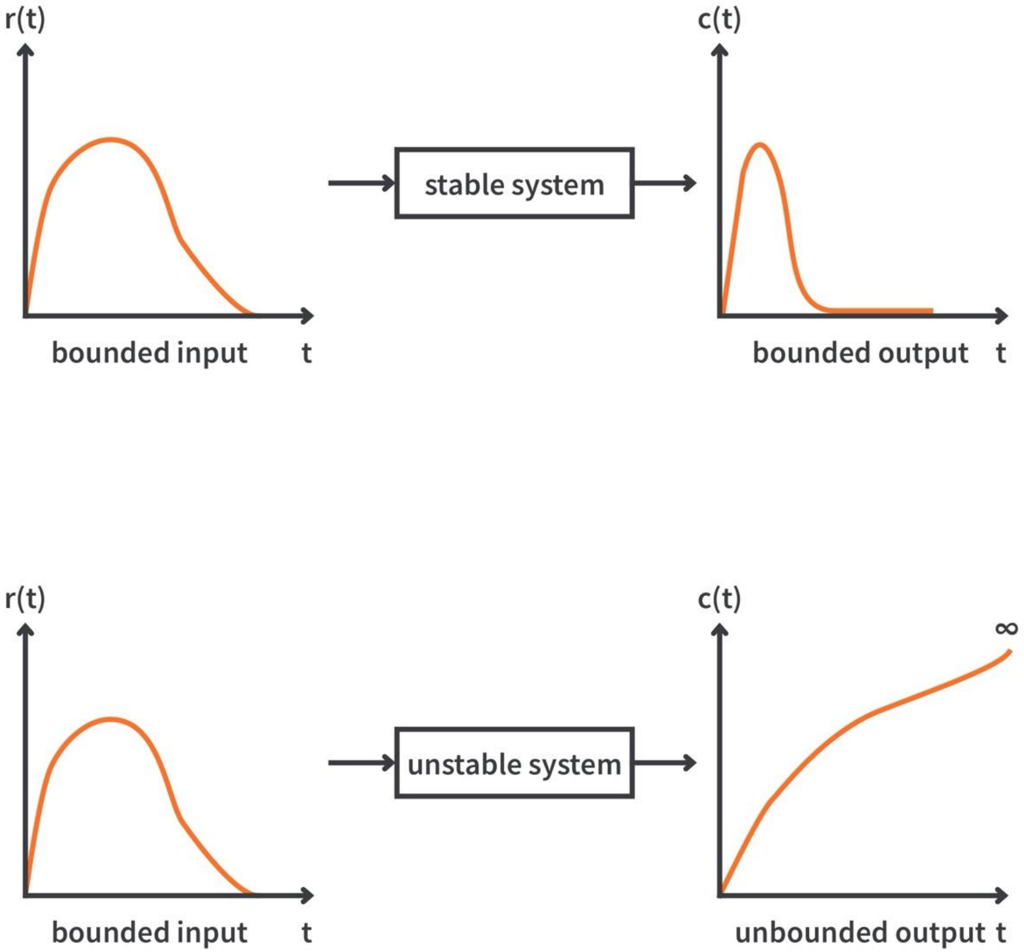
Pole real part determines decay/growth rate

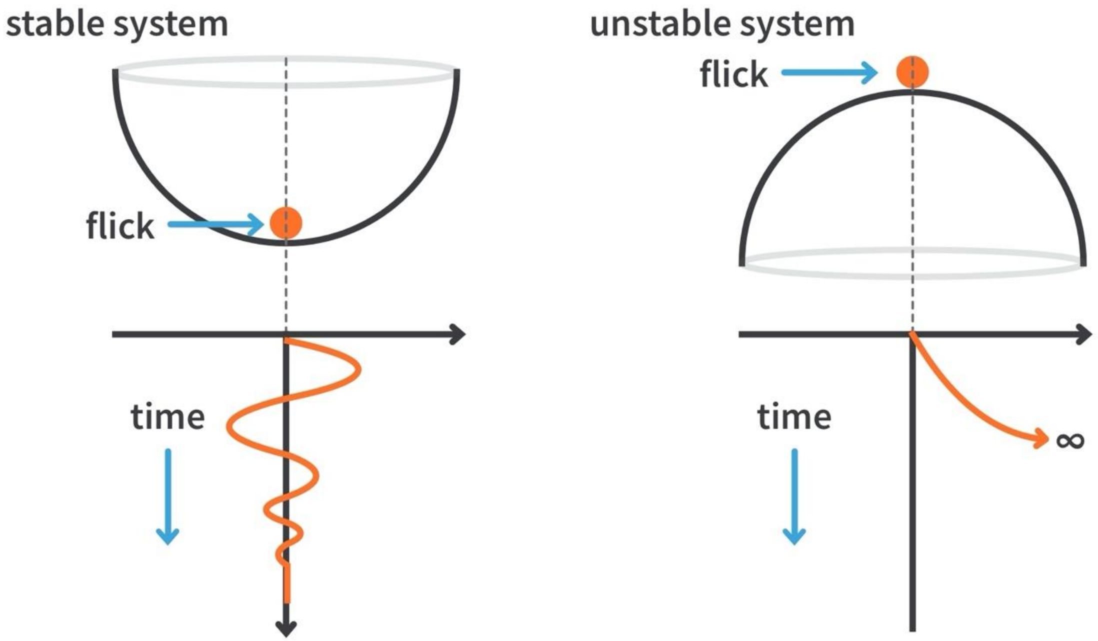
- $\mathrm{Re}(s_p) < 0$, i.e., left half-plane (LHP): exponential decay → stable
- $\mathrm{Re}(s_p) = 0$, i.e., on imaginary axis: purely oscillatory → marginally stable
- $\mathrm{Re}(s_p) > 0$, i.e., right half-plane (RHP): exponential growth → unstable

- Far to the left: fast decay $e^{-\sigma t}$
- Closer to the imaginary axis: slower decay

- Overall transient dominated by pole closest to imaginary axis

Bounded Input Bounded Output (BIBO) Stability





Stability of Continuous-Time Systems

- Stable if all poles have negative real part
- Marginally stable if poles on imaginary axis
- Unstable otherwise

Stability checks for a known fixed transfer function of higher order:

- Numerical solver for root finding: easiest and standard practice
- Analytical alternatives without computing any roots, e.g., Hurwitz criterion (using determinants):
  - Are there any RHP poles?
  - Does stability change with a parameter? (root finding would require parameter scan)

Zero Location

zero at $s = -z$:

$$Y(s) = (s+z)\frac{N_{\text{rest}}(s)}{D(s)}U(s)$$

inverse Laplace transform of $(s+z)U(s)$:

$$\dot{u}(t) + z \cdot u(t)$$

zero at $s = +z$: $\quad \dot{u}(t) - z \cdot u(t)$

Zeros in LHP:

- speed up response (derivative behavior)
- can increase overshoot (caused by complex poles)
- add phase lead (LHP poles have negative phase, so LHP zeros give the smallest possible phase lag → minimum-phase zeros)

$(s+z) \rightarrow j\omega + \omega_z \rightarrow \varphi = \tan^{-1}\left(\frac{\omega}{\omega_z}\right)$

$\omega \to \infty$: $\varphi = +90°$ (LHP pole: $\varphi = -90°$)

Zeros in RHP:

- cause inverse response (initial motion opposite to final steady-state direction)
- add phase lag (→ non-minimum-phase zeros)

- Zeros at the origin introduce differentiating behavior
- Zeros near poles modify or cancel dynamic modes
- An LHP zero close to the imaginary axis or a zero close to a dominant pole (slowest-decaying exponential, i.e., pole with least negative real part) can increase overshoot significantly.

Zeros alone do not make a system unstable, only poles in RHP do.

Interpretation of Zeros

Zeros correspond to multiplications with s → derivatives of the input

- Describe how the input excites the natural modes (described by the poles)
- Creates impulse-like behavior from step ($1/s$) → faster initial rise
- One can cancel a pole with a zero → corresponding exponential term of the pole disappears from the forced response
- Amplification of higher frequencies ($\frac{d}{dt}\sin\omega t = \omega \cdot \cos\omega t$)

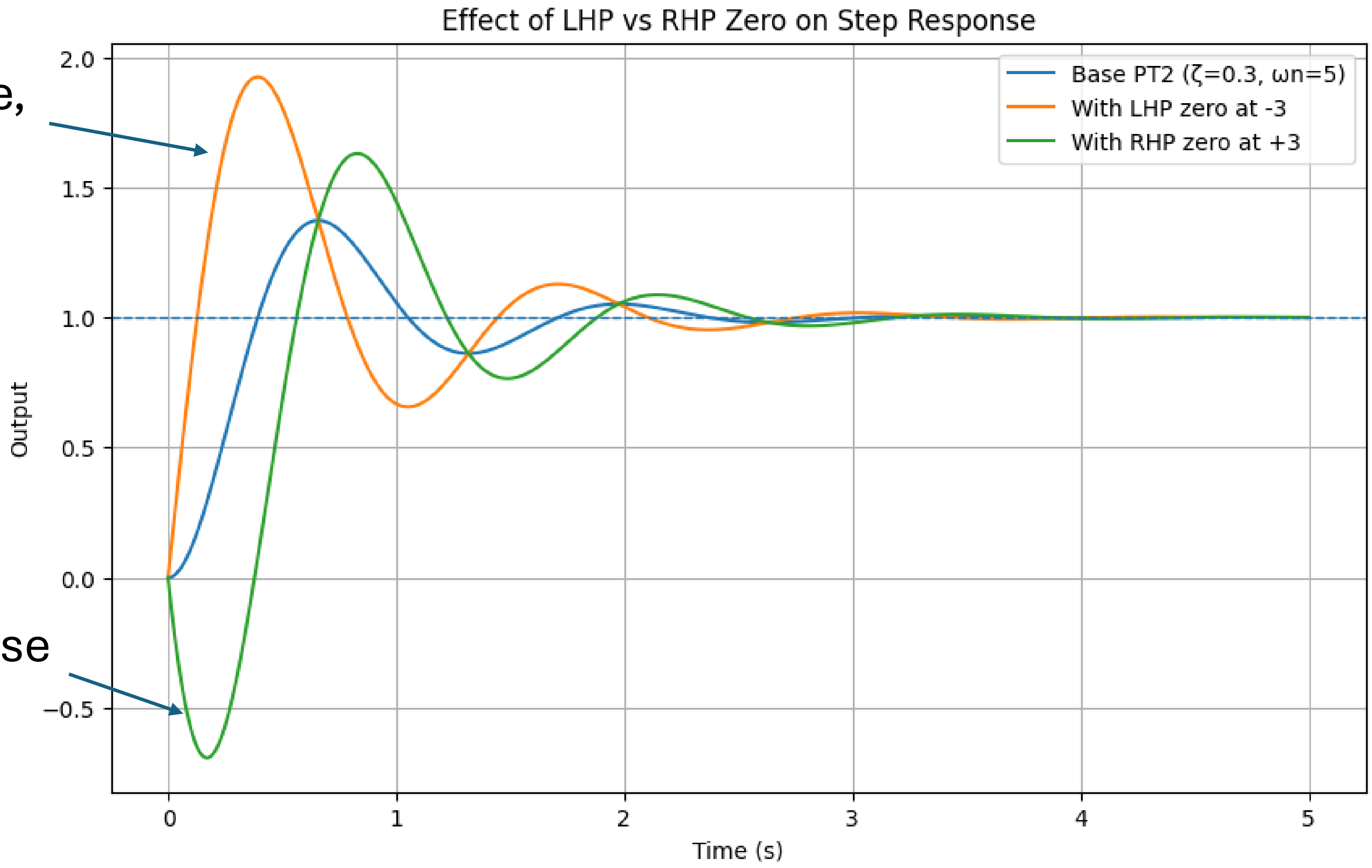
Frequency-selective property: system blocks steady-state sinusoidal response of complex frequency of a zero (despite nonzero input)

More zeros mean a stronger shaping of the input response and an increase of high-frequency gain.

Example: PT2 with LHP and RHP Zero

LHP zero: faster response, increased overshoot

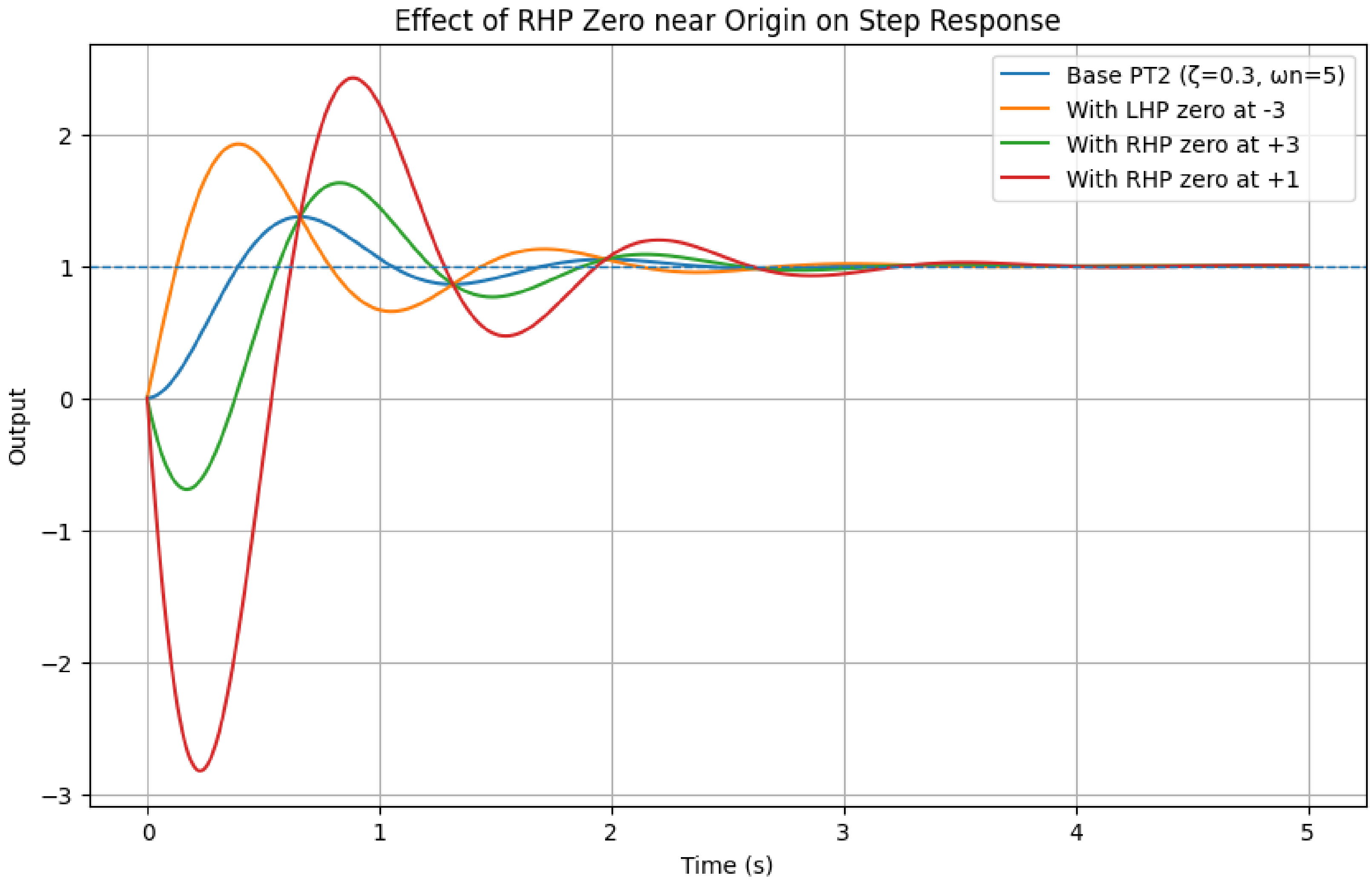
RHP zero: inverse response (initial dip)



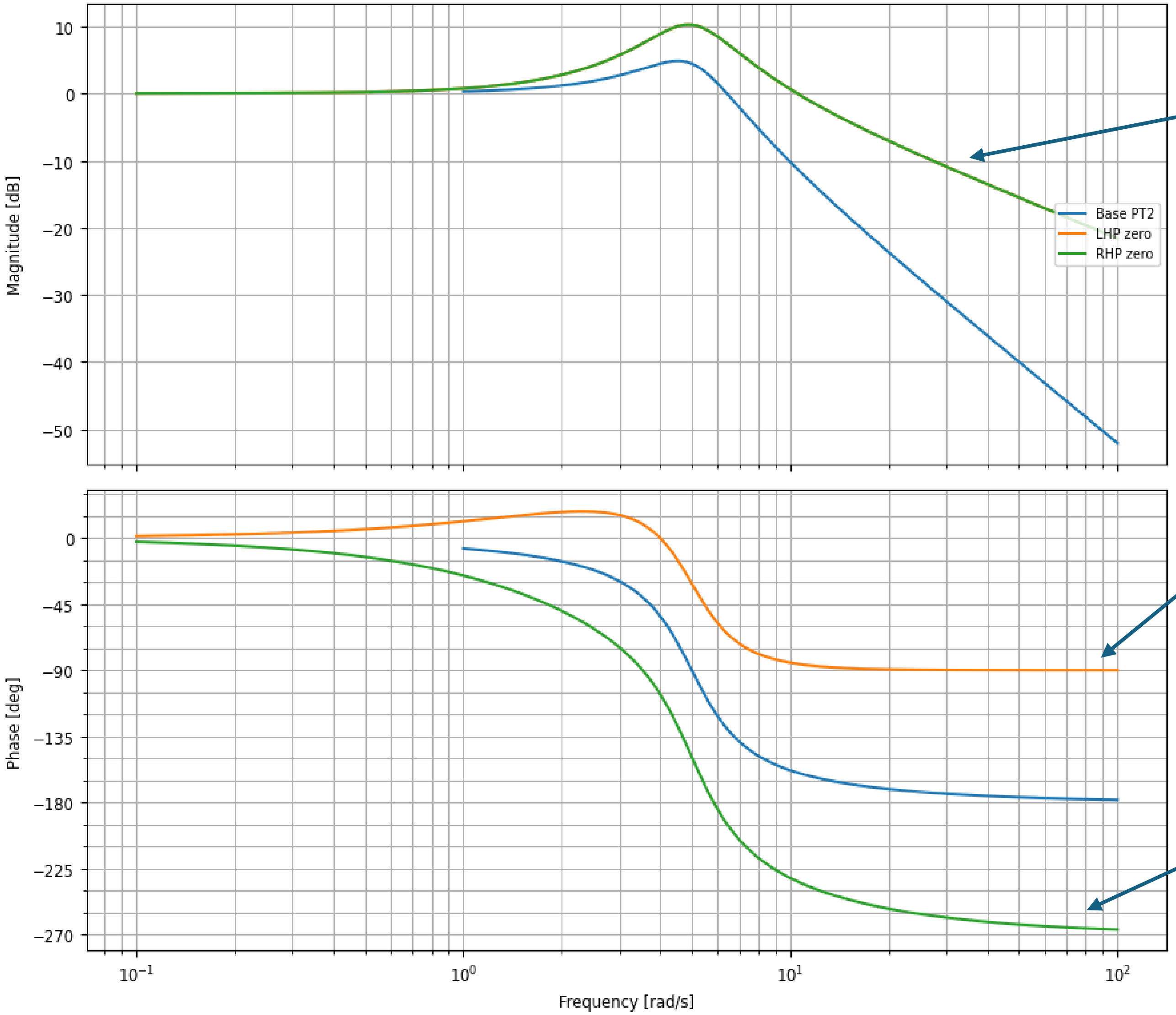
Example: PT2 with LHP and RHP Zero

RHP zero closer to origin:

- stronger dip
- higher overshoot
- more sluggish



Example: PT2 with LHP and RHP Zero



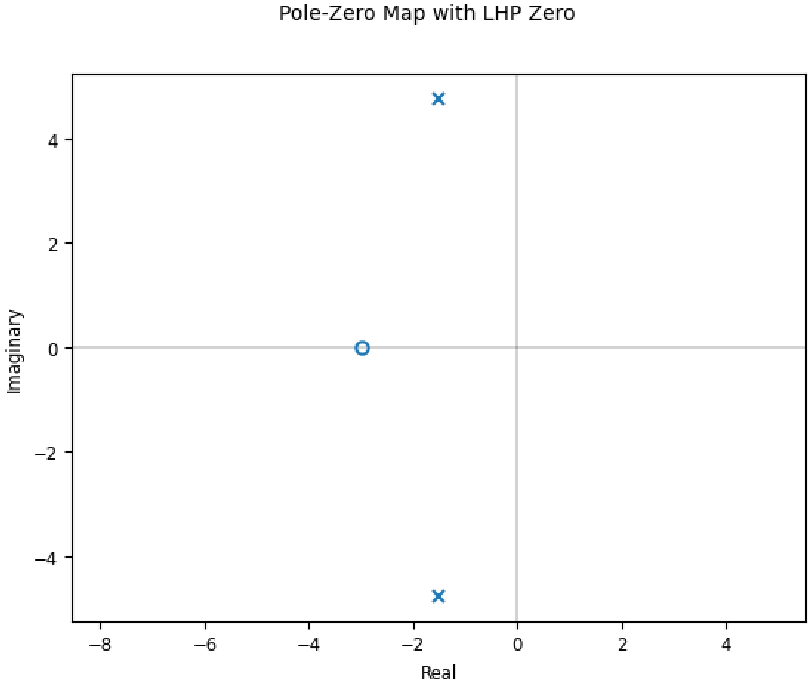
Zeros increase high-frequency slope in the same way

LHP zero: phase lead +90°

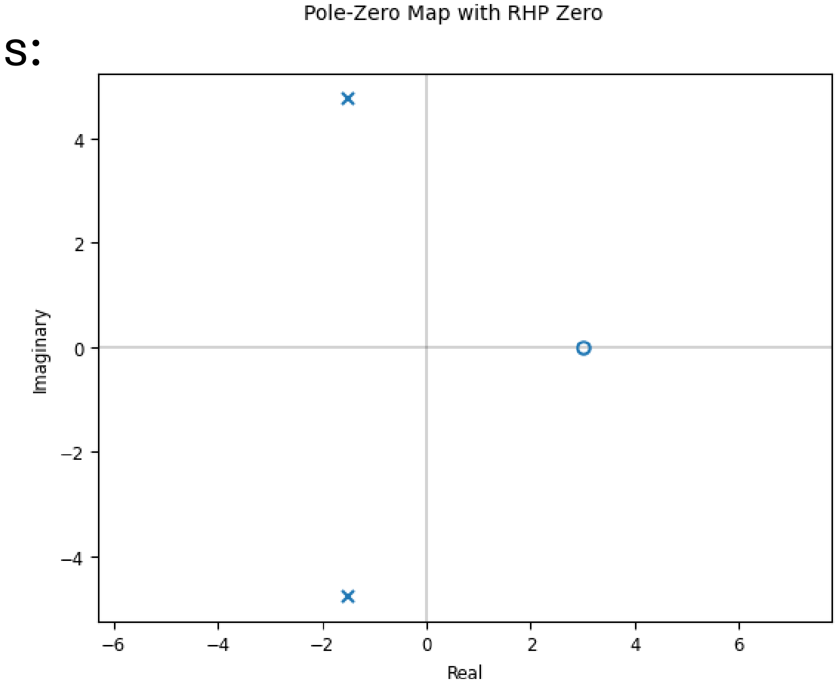
RHP zero: phase lag -90°

Pole-Zero Diagrams

- Graphical representation in complex plane
- Stability visible at a glance
- Often calculated numerically



Example from last slides: both systems stable (poles in LHP)



Resonance

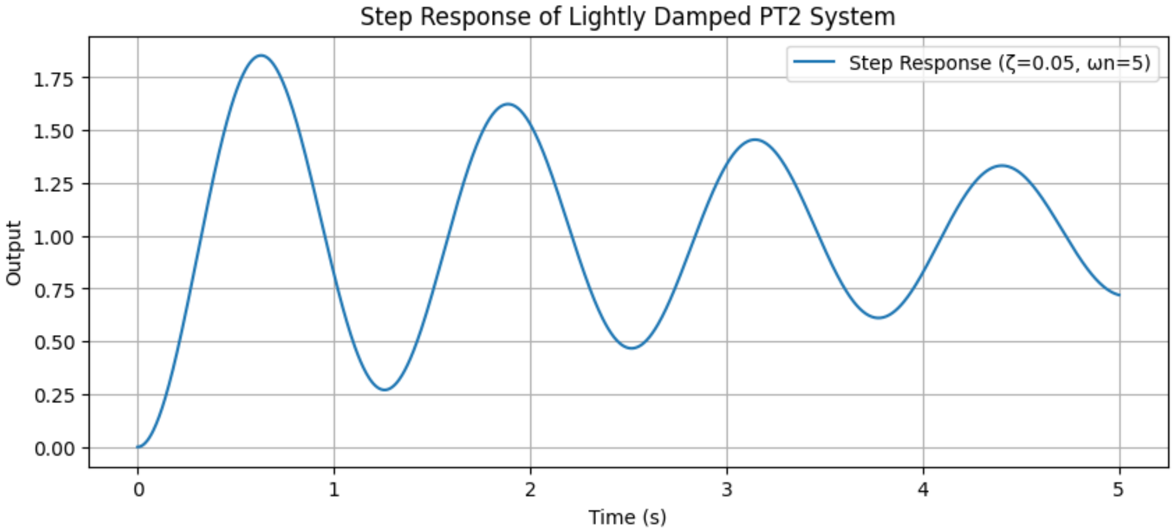
Large response at certain frequencies (if the system's energy storage allows buildup)
→ peaks in frequency response (Bode magnitude, sinusoidal response)

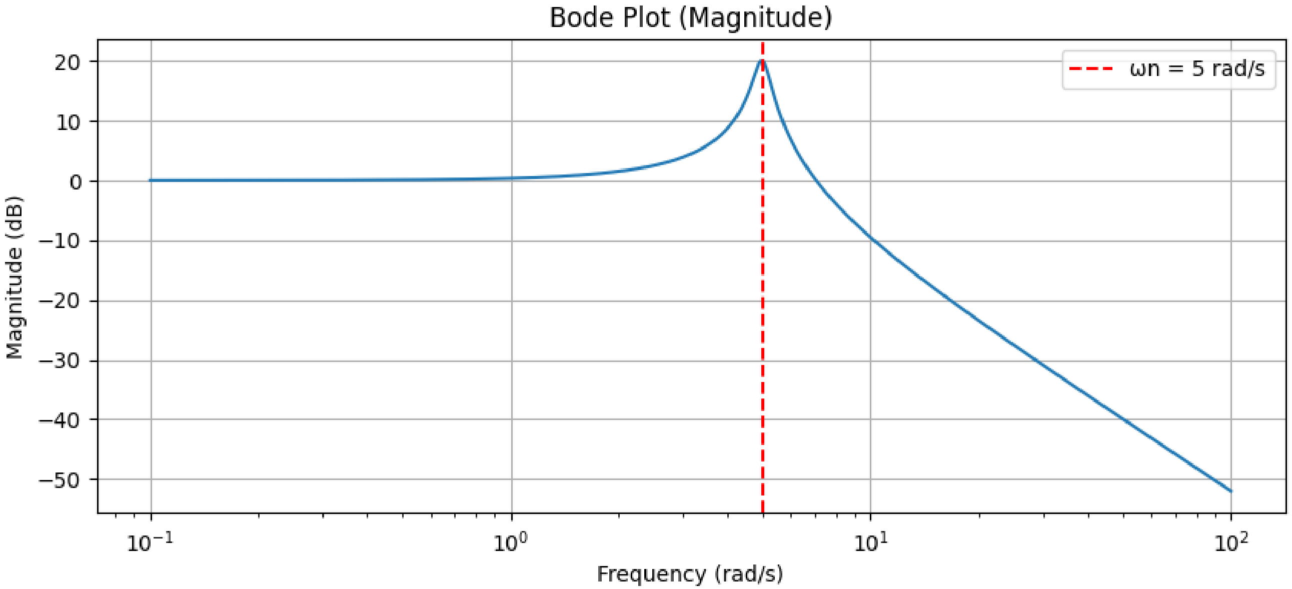
Due to lightly damped complex poles (natural modes)

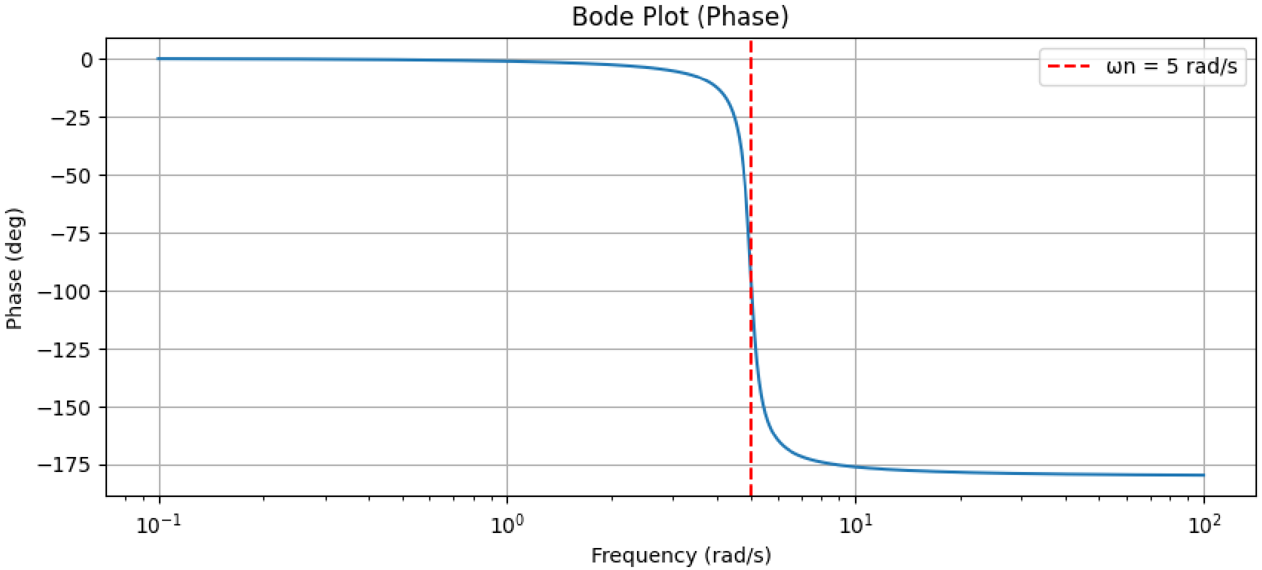
Resonant frequency near imaginary part of the pole
For second-order system: $\omega_r = \omega_n\sqrt{1-2\zeta^2}$ (for small ζ)

Mechanical systems often have very small damping → resonances dominate behavior

→ Controllers must avoid exciting lightly damped poles


Step Response of Lightly Damped PT2 System
Step Response (ζ=0.05, ωn=5)
Output
Time (s)
0
1
2
3
4
5
0.00
0.25
0.50
0.75
1.00
1.25
1.50
1.75


Bode Plot (Magnitude)
ωn = 5 rad/s
Magnitude (dB)
Frequency (rad/s)
20
10
0
−10
−20
−30
−40
−50
10^{-1}
10^{0}
10^{1}
10^{2}


Bode Plot (Phase)
ωn = 5 rad/s
Phase (deg)
Frequency (rad/s)
0
−25
−50
−75
−100
−125
−150
−175
10^{-1}
10^{0}
10^{1}
10^{2}

Frequency-Domain Stability

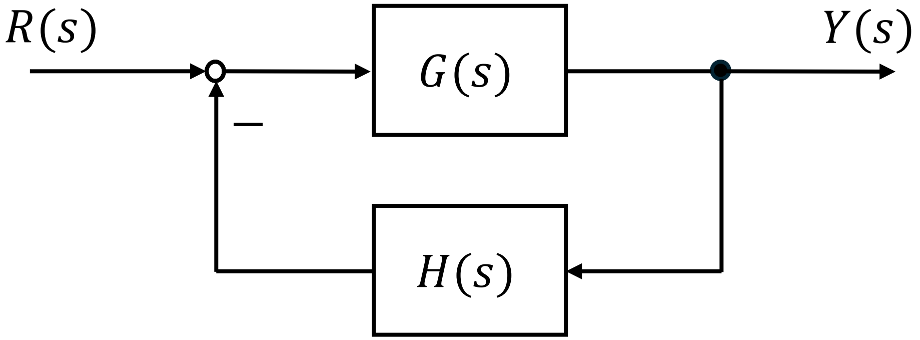
Stability inferred from frequency response, i.e., examination of $G(j\omega)$

- Purely experimental approach, no mathematical model needed, no explicit pole calculation
- Graphical instead of numerical methods: Nyquist curve, Bode plots

So far: open-loop analysis via Bode plots

Now consider closed-loop stability (feedback systems):

Given an open-loop transfer function, is the closed loop stable?



Closed-Loop Stability

Given an open-loop transfer function $G(s)$, the closed-loop transfer function for negative feedback (with feedback path $H(s)$) reads:

$$T(s) = \frac{G(s)}{1+G(s)H(s)} = \frac{\overbrace{G(s)}^{\text{forward path}}}{1+\underbrace{L(s)}_{\text{loop gain}}}$$

Typically, $H(s) = 1$ and a controller $C(s)$ is included in the forward path: $L(s) = C(s) \cdot G(s)$

→ For closed-loop poles: $L(s) = -1$

Unstable if any solution for s lies in RHP

Nyquist Stability Criterion

Consider Nyquist curve for $L(j\omega)$

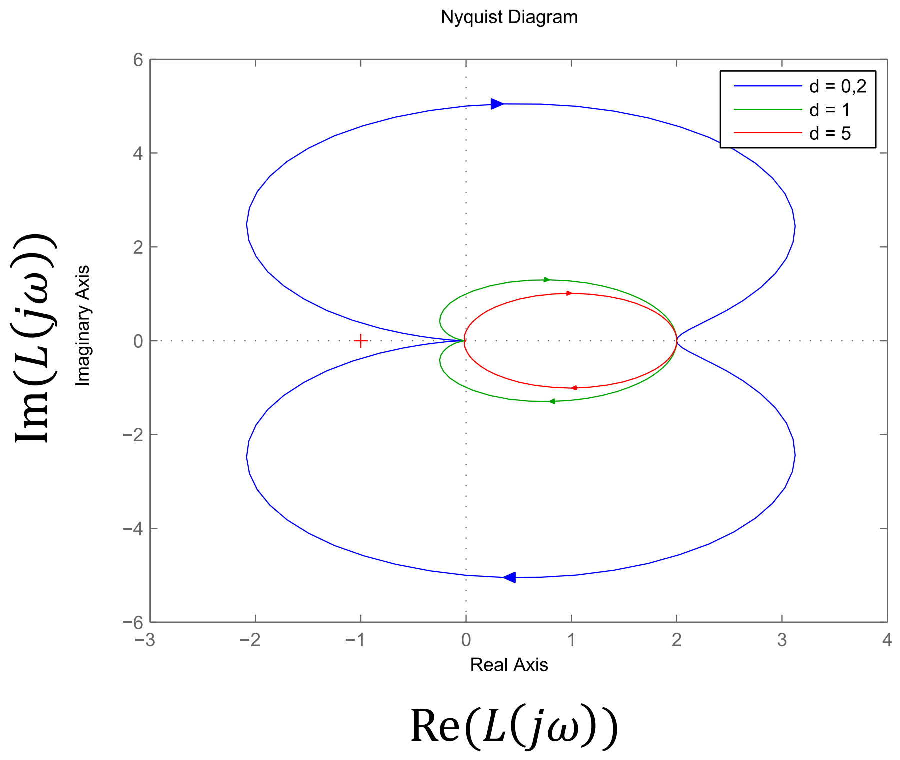
→ evaluation along the imaginary axis of s (boundary of RHP)

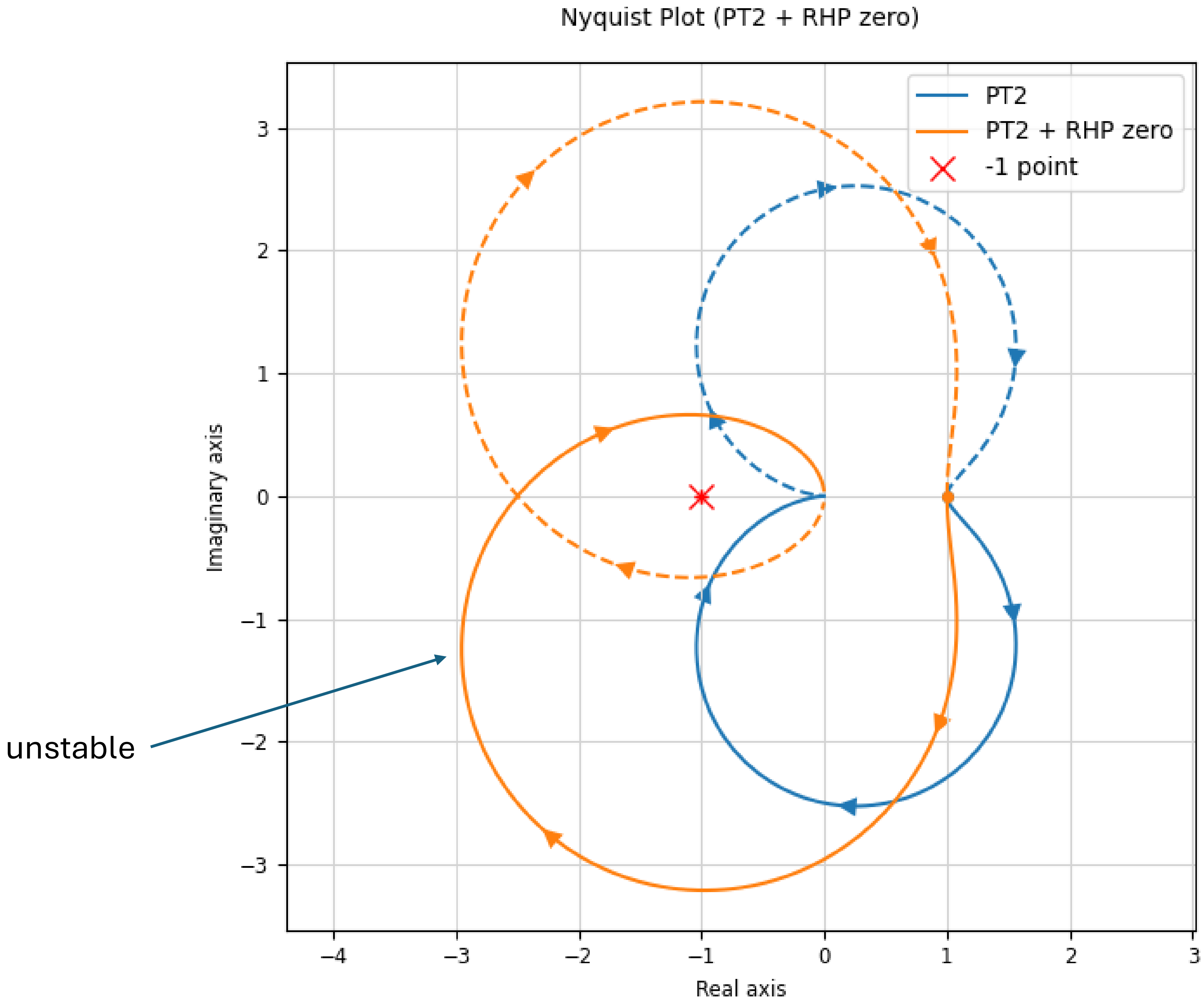
Critical point $-1 + j0$: closed-loop pole

If curve encircles critical point, $L(s) = -1$ has solutions in RHP → closed loop unstable

(Encirclement is a topological signature of a root inside. It considers both positive-frequency and negative-frequency branches: ω: $-\infty \to +\infty$)

Stable PT2 system:





Nyquist Plot (PT2 + RHP zero)
PT2
PT2 + RHP zero
-1 point
Real axis
Imaginary axis
unstable

Nyquist Formula

Number of encirclements → how many closed-loop poles in RHP

- P = number of RHP poles of $L(s)$
- Z = number of closed-loop poles in RHP ($L(s) = -1$)
- N = number of encirclements of -1 by the Nyquist plot of $L(s)$, where the sign reflects the encirclement direction

$$Z = N + P$$

→ An unstable plant (e.g., $P = 1$) can be stabilized by feedback (e.g., $N = -1$).

Gain and Phase Margins

Idea: **measure** distance to instability (of closed-loop system) from Bode plots
→ practical robustness indicators

Instability: $L(j\omega) = -1$
In polar form: $|L(j\omega)| = 1, \quad \angle L(j\omega) = -180°$
- If magnitude is less than 1 at phase $-180°$ → stable
- If magnitude is greater than 1 at phase $-180°$ → unstable

Margins measure distance from that dangerous condition
- Gain margin: how much the loop gain can be increased before instability
- Phase margin: how much more phase lag can be tolerated before instability

Crossover Frequencies

Phase crossover frequency:

$$\angle L(j\omega_{pc}) = -180°$$

Gain margin (unstable when <0):

$$GM = \frac{1}{|L(j\omega_{pc})|}$$

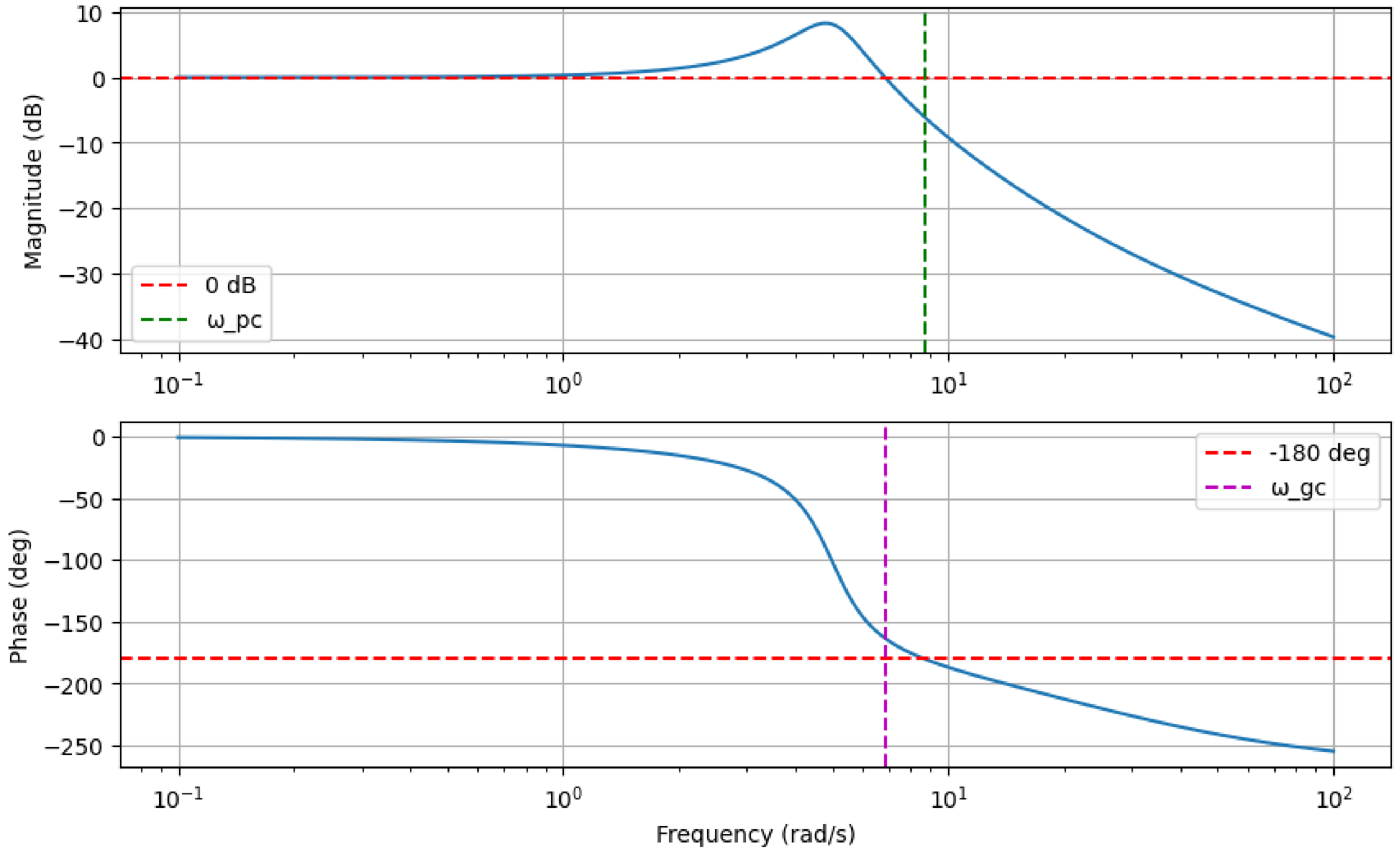
Gain crossover frequency:

$$|L(j\omega_{gc})| = 1$$

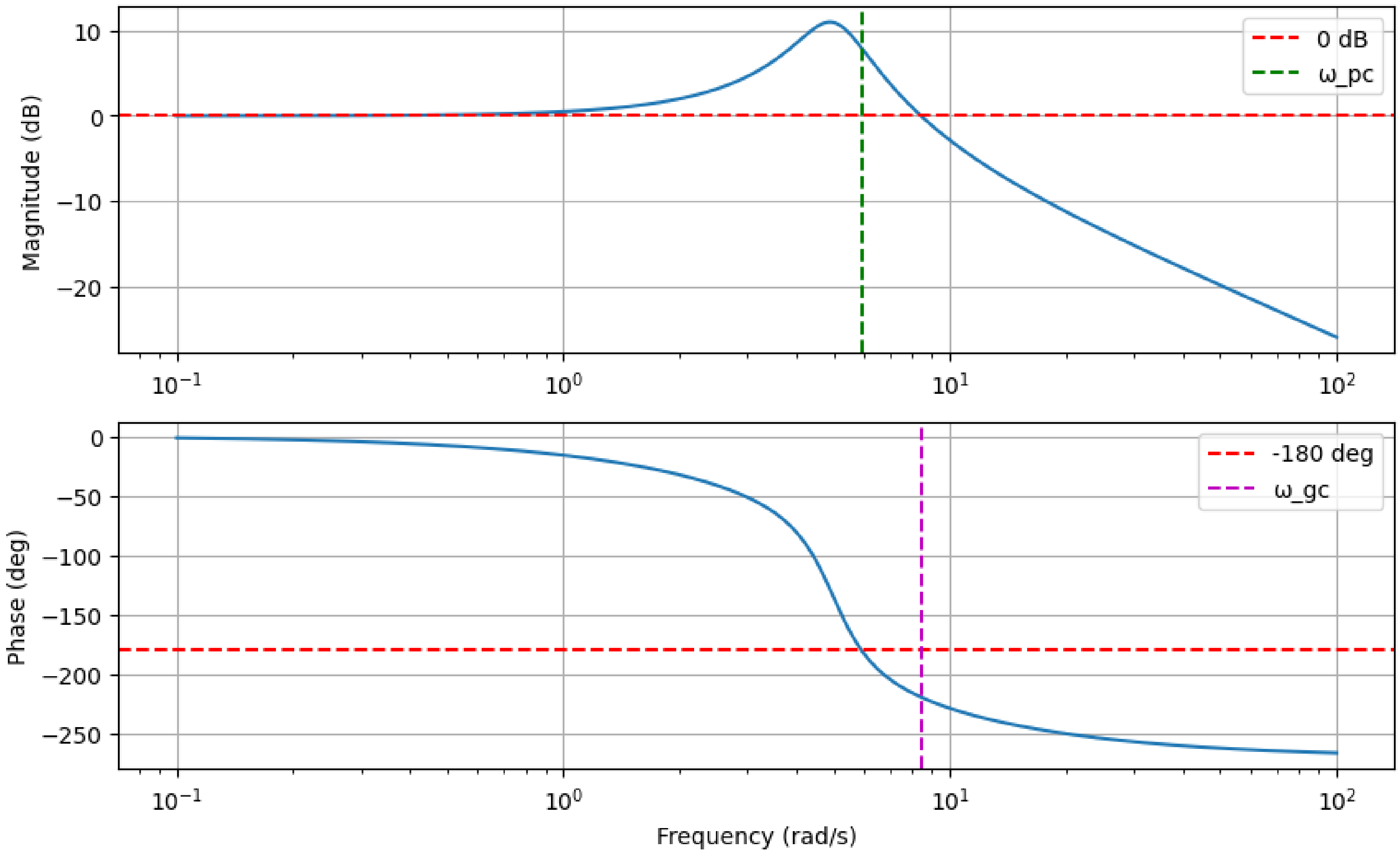
Phase margin (unstable when <0):

$$PM = 180° + \angle L(j\omega_{gc})$$

just barely stable:



unstable:



Magnitude (dB)
10
0
−10
−20
0 dB
ω_pc
Phase (deg)
0
−50
−100
−150
−200
−250
-180 deg
ω_gc
10^{-1}
10^{0}
10^{1}
10^{2}
Frequency (rad/s)

Exercise 1

Consider the transfer function:

$$G(s) = \frac{1}{s^2+4s+5}$$

- Find the poles.
- Determine if the system is stable.
- Classify the damping (overdamped, critically damped, underdamped).
- Predict qualitatively the step response.

Exercise 2

Compare the systems:

$$G_1(s) = \frac{1}{s^2+2s+2}, \quad G_2(s) = \frac{s+1}{s^2+2s+2}$$

- Do both systems have the same poles?
- How does the zero affect the step response?

Exercise 3

For each system, determine stability:

$$G_1(s) = \frac{s+1}{(s+2)(s+3)}$$

$$G_2(s) = \frac{1}{(s-1)(s+2)}$$

$$G_3(s) = \frac{1}{s^2+2s+1}$$

Exercise 4

Given poles:

- $s = -1 \pm j2$
- $s = -10$

- Which poles dominate?
- Describe the system behavior.

Exercise 5

Consider the following two systems:

$$G_1(s) = \frac{1}{(s+2)(s+4)}, \qquad G_2(s) = \frac{s-1}{(s+2)(s+4)}$$

- Determine the poles and zeros of both systems.
- Compare stability.
- Predict how the zero in $G_2(s)$ affects the step response.
- What is special about the zero location in $G_2(s)$?

Exercise 6

Given:

$$G(s) = \frac{\omega_n^2}{s^2+2\zeta\omega_n s+\omega_n^2}$$

Let $\omega_n = 10, \zeta = 0.2$

- Is resonance expected?
- Estimate if the peak is strong or weak.

Exercise 7

You are given:

- Gain crossover frequency: $\omega_{gc} = 5\,\mathrm{rad}/s$
- Phase at crossover: $-150°$

- Compute the phase margin.
- Determine stability.